

Aryan Panchal

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PERSONAL STATEMENT

An aspiring engineer who has proven critical thinking skills through competitive STEM activities like Science Olympiad and Chess at national title games. Enthusiastic about applying technical expertise to real-world challenges in IT or development.

EDUCATION

East Stroudsburg University of Pennsylvania

Bachelor of Science Computer Science (Dean's List) (GPA: 3.8)

East Stroudsburg, PA

May 2027

WORK EXPERIENCE

East Stroudsburg University of Pennsylvania

IT Technician

East Stroudsburg, PA

December 2024 - Present

- Diagnosed and fixed hardware, software, and network issues
- Delivered IT support via phone, email, or in-person to university staff or students
- Systemized set-up and maintenance of university computers and related equipment

PROJECT EXPERIENCE

2D Java Game Engine

December 2023 – Present

- Hand drew all game assets, i.e., player animations each frame, terrains, etc
- Implemented game loops, entity management, and input handling from scratch
- Custom collision detection systems for player-environment interactions
- Utilized Java's Graphics2D for sprite animation and scene management
- Optimized performance via efficient garbage collection and rendering techniques

Autonomous Snake Game

March 2025 - Present

- Developed in Unity 3D using C# for scripting
- Implementation of A* pathfinding to calculate shortest path to food while dynamically avoiding collisions
- A* makes use of Manhattan/Euclidean distance heuristics
- Utilized priority queues and efficient data structures to manage larger grids smoothly

SKILLS

- **Languages:** Java, C++, Python, JavaScript, C#, C, Assembly, Lisp
- **Concepts:** Object-Oriented Programming (OOP), Data Structures, Algorithms, Software Development Life Cycle (SDLC), Agile Methodologies, Version Control
- **Core Courses:** Computer Organization, Computer Architecture, Data Structures, Algorithms, Cybersecurity, Programming Languages, Operating Systems, Software Engineering
- **Mathematic Disciplinary Courses:** Calculus I, Calculus II, Discrete Mathematics, Statistics: Probability Theory, Linear Algebra
- **Major Electives:** Artificial Intelligence, Computer Graphics